



Climbing the Climate Staircase: How Natural Cycles and Greenhouse Gases Shape Global Temperatures

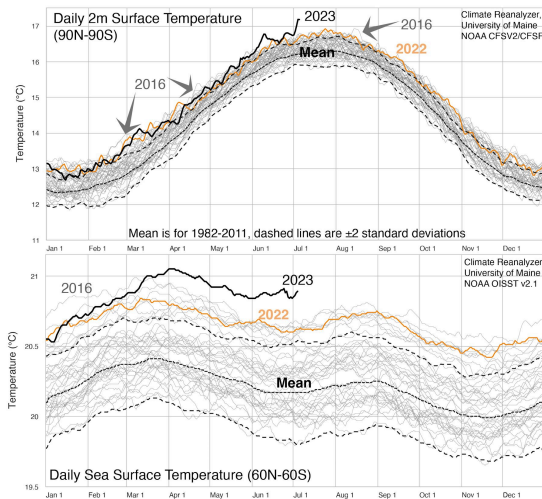
Abstract

In mid-2023, climate records were falling everywhere: the highest daily global temperatures ever measured, record sea surface temperatures, and record-low Antarctic sea ice. Climate scientist Kevin Trenberth explained that global temperatures climb in steps, usually at the end of El Niño events, and predicted that 2023 and 2024 would take the next big step up the staircase.

Introduction

Global warming took off in the mid-1970s, when the rise in global temperature exceeded natural variability. Every decade since the 1960s has been warmer than the one before, and the 2010s were the warmest on record. But from year to year, temperatures bounce around. In early July 2023, Earth recorded its highest daily temperatures ever, alongside the largest sea surface temperature anomaly ever. June had its highest global mean surface temperature, and Antarctic sea ice hit a record low. Carbon dioxide kept rising with no sign of slowing, while some nations continued building coal plants whose emissions more than offset cuts elsewhere. Why were so many records breaking at once?

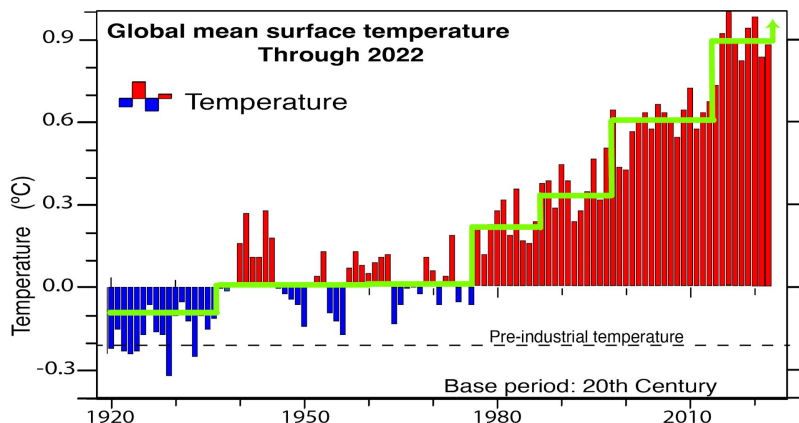
How Do Scientists Study This?



FROM THE ARTICLE

Daily global surface air temperatures (top) and sea surface temperatures (bottom) breaking records in 2023. Author provided, based on NOAA analyses processed by the University of Maine, CC BY-SA.

Trenberth analyzed NOAA's global temperature record since 1920 and fitted it with steps rather than a straight line, showing when temperatures jumped to new levels. He compared daily global surface air and sea surface temperatures for 2023 with every year since 1982, using data processed by the University of Maine. He linked each step to natural variability: El Niño events, which move heat from the ocean into the atmosphere, and the Pacific Decadal Oscillation, which controls whether extra heat is stored in the upper 300 meters of the ocean or buried deeper. He also used ocean heat content, which rises more steadily than air temperature, as a check on long-term warming.

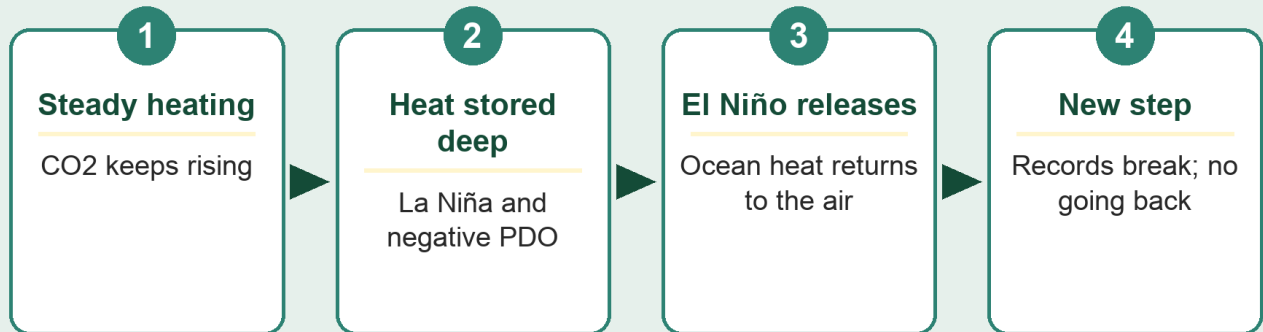


FROM THE ARTICLE

Global temperatures climb in steps, usually at the end of an El Niño event. Author provided, based on NOAA data, CC BY-SA.

Explore the Evidence

How the climate staircase forms



Fit the steps

Global temperatures since 1920 were grouped into flat periods and sudden jumps.

Match to El Niño

Scientists checked whether jumps followed the end of El Niño events.

Check ocean heat

Steadily rising ocean heat confirms that warming never paused.

What Did They Discover?

The temperature record looks like a rising staircase. The warmest 20th-century year was 1998, after a major El Niño; then came the 2001-2014 hiatus, which deniers used to claim warming was a myth. The 2015-16 El Niño ended that: 2015 set a record, beaten by 2016. During El Niño, heat stored deep in the western Pacific returns to the atmosphere, causing a mini global warming. In 2023, sea surface temperatures became the highest on record in April, running 0.2°C above previous highs, setting up record air temperatures in June and July. Trenberth predicted 2023 would be the warmest year to date and 2024 could step up to about 1.4°C above pre-industrial levels, with daily incursions above 1.5°C.

2023's record-breaking year

+0.2°C sea surface temperatures above previous record highs

~1.4°C predicted 2024 warming above pre-industrial

2016 previous warmest year, after a major El Niño

After the next La Niña, the rise will pause again, but temperatures will never quite return to previous levels.

Takeaways

A staircase, not a ramp

Natural cycles plus steady heating make temperatures rise in steps.

El Niño takes the step

The biggest jumps come late in El Niño events as ocean heat returns to the air.

Ocean heat tells the truth

Ocean heat rises steadily, showing warming continues even during pauses.

Key Terms

Anomaly

A departure from the long-term average, such as how much warmer the sea is than normal.

Pacific Decadal Oscillation

A decades-long pattern in the Pacific that affects how much heat is stored deep in the ocean.

Hiatus

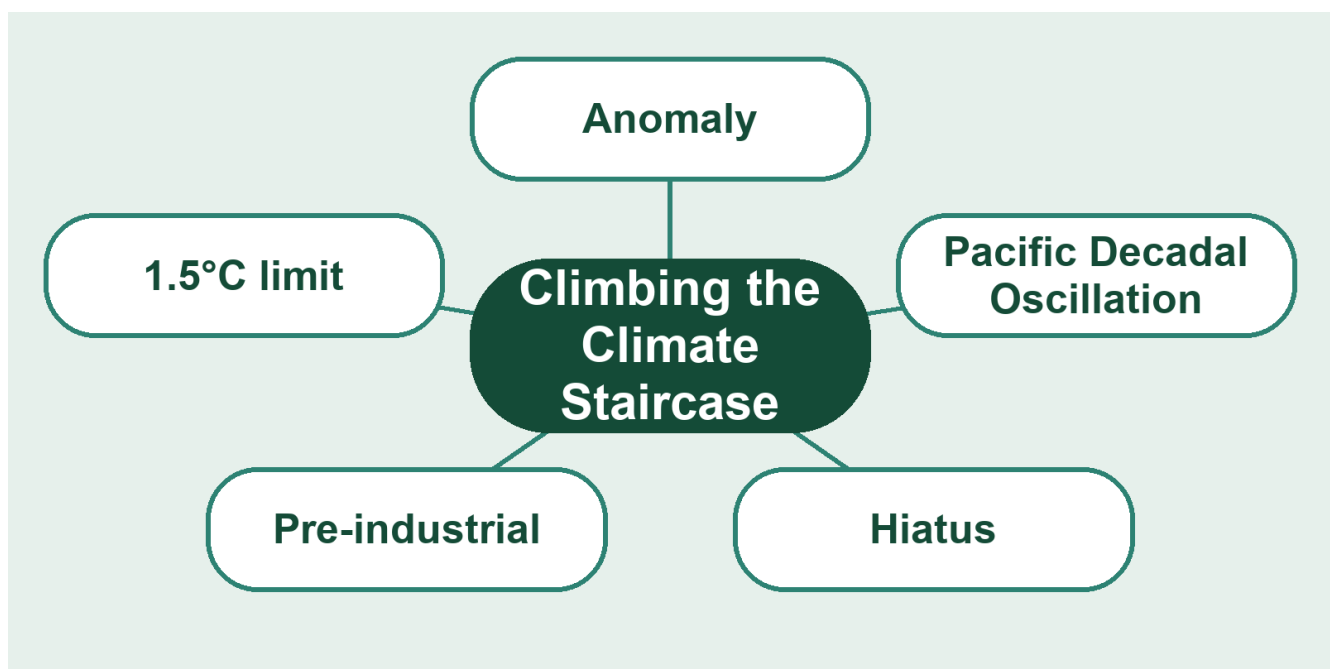
A period when global surface temperature rose more slowly, even though Earth kept gaining heat.

Pre-industrial

The period before widespread fossil fuel use, used as a baseline for measuring warming.

1.5°C limit

The warming level the Paris Agreement aims to stay below to avoid the worst impacts.



Activity

Predict the Next Step

Students analyze the staircase pattern and make an evidence-based prediction about future temperatures.

Materials: Printed staircase temperature graph from the lesson; Rulers; Colored pencils

1. On the graph, trace each flat step and mark the years when temperatures jumped.
2. Label which jumps followed major El Niño events, such as 1997-98 and 2015-16.
3. Estimate the average height of each step.
4. Using Trenberth's reasoning, sketch where the next step after 2023-24 might be.
5. Write a paragraph explaining why a cool year after an El Niño does not mean warming has ended.

References

Trenberth, K. (2023). Global temperature rises in steps: here's why we can expect a steep climb this year and next. The Conversation. <https://doi.org/10.64628/AA.75nadgn7u>